

Problem Set 5

Date _____ 20 ____

Section 4.1

7) $y_1 = y_2 = 1$ and $y_3 = -1$. Then we have:-

$$\begin{array}{rcl} 1 \times & x_1 - x_2 & = 1 \\ 1 \times & x_2 - x_3 & = 1 \\ -1 \times & x_1 - x_3 & = 1 \end{array}$$

$$\begin{array}{rcl} x_1 - x_2 + x_2 - x_3 - x_1 + x_3 & = & 1 + 1 - 1 \\ 0 & = & 1 \end{array}$$

9) Blank 1:- Column Space
Blank 2:- Orthogonal

31) B is a basis for the row space of A .

~~32) Assuming the matrix has rank 1 (because otherwise either both column spaces or both nullspaces will be 0 and~~

~~a) A~~
32) Assuming the matrix has rank 1, for κ and l to be a basis for $C(A)$ and $N(A^T)$ we need them to be orthogonal i.e. $\kappa \cdot l = 0$. For r and h to be a basis for $C(A^T)$



and $N(A)$ we need r and n to be orthogonal i.e. $r \cdot n = 0$.

b) Cr^T

33) a) Assuming rank=2:-

1) ~~We~~ Each pair of vectors must be independent (e.g. $r_1 \neq \alpha r_2 \forall \alpha \in \mathbb{R}$)

~~The vectors~~ We must have $r_1 \perp n_1$, $r_1 \perp n_2$, $r_2 \perp n_1$, $r_2 \perp n_2$ and we must have $c_1 \perp l_1$, $c_1 \perp l_2$, $c_2 \perp l_1$, $c_2 \perp l_2$.

b) $A = (c_1 \ c_2) (r_1 \ r_2)^T$

Section 4.2

13) $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$P = A(A^T A)^{-1} A^T$

P has shape $\cancel{3 \times 3}$
 4×4



$$A^T A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(A^T A)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{Because } I^{-1} = I)$$

$$\cancel{A} A (A^T A)^{-1} A^T =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = P$$

Projection of $b = Pb$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}$$

16) $Ax = b \therefore \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b \therefore \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\hat{x}_1 = 1/2 \quad \text{and} \quad \hat{x}_2 = 3/2$$



So the closest linear combination to b is:-

$$\hat{\gamma}_1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + \hat{\gamma}_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \text{ where } \hat{\gamma}_1 = 1/2 \text{ and } \hat{\gamma}_2 = 3/2$$

Since b was already in the span of A , its projection onto $C(A)$ was also b therefore the above combination gives b itself.

$$\begin{aligned} 17) \text{ Suppose } P^2 &= P. \text{ Then } (I-P)^2 = \\ (I-P)(I-P) &= I^2 - PI - PI + P^2 \\ &= I - P - P + P \\ &= I - P. \end{aligned}$$

First blank:- Left nullspace of A .

30) a) The matrix has rank 1 so the column space is spanned by all multiples of the vector $a = \begin{bmatrix} 3 \\ 4 \end{bmatrix}^T$. Then compute P_c as follows:-

$$P_c = \frac{a a^T}{a^T a} = \frac{\begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix}}{\begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}} = \frac{\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}}{25}$$



b) Row space of $A = C(A^T)$

$$= C \left(\begin{bmatrix} 3 & 4 \\ 6 & 8 \\ 6 & 8 \end{bmatrix} \right)$$

Since A^T has rank 1, this we let $a_r = [3 \ 6 \ 6]^T$. Then P_r can be calculated:-

$$\frac{a_r a_r^T}{a_r^T a_r}$$

$$= \frac{\begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix} [3 \ 6 \ 6]}{\begin{bmatrix} 3 & 6 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix}}$$

$$= \frac{\begin{bmatrix} 9 & 18 & 18 \\ 18 & 36 & 36 \\ 18 & 36 & 36 \end{bmatrix}}{81}$$

$$= \begin{bmatrix} 1/9 & 2/9 & 2/9 \\ 2/9 & 4/9 & 4/9 \\ 2/9 & 4/9 & 4/9 \end{bmatrix}$$

$$B = P_C A P_R \Leftrightarrow$$

$$= \begin{bmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{bmatrix} \begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix} \begin{bmatrix} 1/9 & 2/9 & 2/9 \\ 2/9 & 4/9 & 4/9 \\ 2/9 & 4/9 & 4/9 \end{bmatrix}$$

$$= \begin{bmatrix} 27 \\ 48 \end{bmatrix} \begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix} \begin{bmatrix} 1/9 & 2/9 & 2/9 \\ 2/9 & 4/9 & 4/9 \\ 2/9 & 4/9 & 4/9 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$$

$B = A$ because P_C projects vectors onto $C(A)$, and so $P_C A = A$ since it is simply projecting A 's columns onto $C(A)$. Also $A P_R = A$ because P_R is projecting the rows of A onto its row space. Also, for the same reason, $P_R A^T = A^T$
 $(P_R A^T)^T = A$
 $A P_R = A$

Therefore $P_C A P_R = A$.

31) Since P Let the matrix $A = (a_1, a_2, \dots, a_n)$. Then the projection matrix P is $A(A^T A)^{-1} A^T$. If $Pb = p$, then p is the projection of b onto the subspace spanned by the a 's.



31) Let the matrix $A = (a_1, a_2, \dots, a_{n-1}, a_n)$.
 Then ~~if~~ let $e = b - p$. If $A^T e = 0$
 the p is the projection of
 b onto $C(A)$.

34) Let B be an $\overset{r \times n}{\uparrow}$ matrix with
 independent rows.

Then B^T is $n \times r$ with r
 independent columns. We know
 from the question that $A^T A$ is
 invertible when A is $m \times r$ with
 independent columns, so it follows
 that $B B^T$ is invertible.

Let A be an $m \times r$ matrix with
 r independent columns and let
 B be an $r \times n$ matrix with
 r independent rows.

$A^T A$ is $r \times r$ and as show, is invertible
 so it has full row & column rank i.e r ,
 and same for $B B^T$, and so $A^T A B B^T$
 is $r \times r$ with rank r . If we look
 Also, since A^T multiplication by A^T and
 B^T cannot increase rank, $A^T A B$ has
 the same rank as $A B$ and
 $A B B^T$ has same rank as



AB , thus $A^T A B B^T$ has same rank as AB (i.e. rank r)

~~Section 8.2~~